Karl Jaehnig

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Work Experience

AI / ML Instructor – *AI-4-ALL* **January 2024 – Present**

- Mentored and guided cross-functional student teams to design, train, and launch analytics pipelines that cleaned, integrated, and visualized multi-modal datasets within a 13 week spring cycle.
- Revamped Colab-based coding modules to emphasize data extraction, data transformation, and data visualizations to boost student engagement.
- Engineered interactive Plotly dashboards in Streamlit to translate student survey results into data driven insights that informed iterative improvements to AI-4-ALL programs.

Data Analyst – *Center for Computational Astrophysics, Flatiron Institute* **August 2021 – January 2022**

- Developed a Bayesian hierarchical model in PyMC3 that achieved sub-3% error for robust parameter estimation for complex highly variable time-series datasets.
- Employed Gaussian Process models using Celerite to denoise multi-modal time-series datasets to enhance signal clarity for forecasting analysis.
- Optimized Python code to create a lightweight Matplotlib application that visualized metric evolution over time to simplify complex concepts for non-technical stakeholders.

Researcher – *Department of Physics & Astronomy, Vanderbilt University* **January 2019 – August 2023**

- Published several peer-reviewed manuscripts on using unsupervised machine-learning frameworks to create clean, analysis-ready data catalogs.
- Constructed SQL - Python pipelines to extract, transform, and load multi-sourced astronomical datasets to create a unified, queryable database.
- Increased pattern detection sensitivity by 11% through probabilistic clustering algorithms with Scikit-Learn within large, noisy datasets.

Projects

Fraudulent Credit Card Transaction Detection **September 2025**

- Conducted Exploratory Data Analysis using PySpark to explore trends within a credit card dataset containing over 24M transactions across three databases to uncover trends driving fraudulent activity.
- Constructed pipeline to handle Data Pre-processing, Feature Engineering, and Random Forest modeling with weighted classes to correct for extreme imbalances in transaction type.
- Optimized fraud model performance via hyper-parameter tuning and distributed compute settings to achieve a recall of 90.8% and precision of 91.7% when flagging anomalous transactions.

Probabilistic Pneumonia Detection in X-Rays **July 2024**

- Accomplished feature extraction from pediatric chest x-rays using an EfficientNetV2S pre-trained convolutional neural network in TensorFlow Applications.
- Utilized a probabilistic model to distinguish between bacterial and viral pneumonia with full treatment of uncertainty with TensorFlow-Probability library.
- Achieved an F1-Score of 96.8% with a sensitivity of 97.7% and specificity of 92.4% after fine-tuning hyper-parameters using Bayesian Optimization.

Interactive arXiv Paper Explorer **May 2024**

- Developed XML parsers to extract identifying information related to user-query related manuscripts within the open access archive database arXiv in Python3.
- Performed Natural Language Processing of downloaded manuscript abstracts using NLTK to remove stop-words and calculate Semantic Similarity between manuscript abstracts to highlight subject overlap.
- Deployed a Streamlit web application to visualize arXiv user search results as interactive Network Graphs created using PyVis to convey both abstract semantic similarity and author overlap.

Technical Skills

Programming	Python, SQL, Rust
ML Libraries	NLTK, Open-CV, TensorFlow, XML, Scikit-Learn, Pandas, TensorFlow, Keras, PyTorch
Analytics	Bayesian statistics, Exploratory Data Analysis, Forecasting, Time-series analysis
Big Data / Cloud	Apache Spark, GCP (BigQuery)
Dev / MLOps	Docker, Git, Streamlit, Jupyter, Colab, MLFlow
Visualizations	Excel, Matplotlib, Plotly, Dash, Seaborn, Tableau

Education

- **Doctorate** – *Astrophysics, Vanderbilt University* **August 2023**
- **Master of Science –** *Physics, Fisk University* **July 2017**
- **Bachelor of Science** *Astronomy, University of Florida* **December 2011**